

MDSPACE Installation Guide

Enterprise Linux repositories, portable Linux bundles, and AlmaLinux virtual machines

Supported CPU architectures: x86_64 and aarch64

Before you begin: For testing, the recommended installation method is the portable Linux bundle. On Windows and macOS, create an AlmaLinux 9 virtual machine and run the EL9 bundle inside it. Prebuilt virtual machines are also provided and can be imported as an alternative. All installation assets are available at <https://gallois.cc/assets/cryonet.html>.

1. Choose an installation method

Host system	Recommended method	Architecture	Notes
AlmaLinux/RHEL/Rocky 9	EL9 RPM packages	x86_64 or aarch64	Native installation with dnf.
AlmaLinux/RHEL/Rocky 10	EL10 RPM packages	x86_64 or aarch64	Native installation with dnf.
Other Linux, glibc 2.34–2.38	EL9 portable bundle	x86_64 or aarch64	Self-contained bundle with external tools.
Other Linux, glibc 2.39 or newer	EL10 portable bundle	x86_64 or aarch64	Preferred bundle for newer systems.
Linux with glibc older than 2.34	Virtual machine	Match host CPU	Portable bundles are not supported.
Windows	AlmaLinux 9 VM + EL9 bundle	Match host CPU	Create an AlmaLinux 9 VM, or import the supplied prebuilt VM, then run the EL9 bundle.
macOS	AlmaLinux 9 VM + EL9 bundle	Match host CPU	Create an AlmaLinux 9 VM in UTM, or import the supplied prebuilt VM, then run the EL9 bundle.

2. Identify your CPU architecture

On Linux, run:

```
uname -m
```

Interpret the result as follows:

Output	Download architecture
x86_64	x86_64
aarch64 or arm64	aarch64

Important: Packages, bundles, and virtual appliances are architecture-specific. An x86_64 build

cannot run natively on an aarch64 system, and an aarch64 build cannot run on x86_64.

3. Native installation on Enterprise Linux 9 or 10

This method is intended for AlmaLinux, Red Hat Enterprise Linux, Rocky Linux, and compatible distributions. The external MDSPACE dependencies are distributed as binary RPM assets in the continuous GitHub release. Install the matching EL9 or EL10 RPMs directly with dnf, together with the separately supplied MDSPACE RPM.

3.1 Confirm the Enterprise Linux major version

```
cat /etc/os-release
rpm -E %rhel
```

Use the EL9 packages when the major version is 9 and the EL10 packages when it is 10. Do not mix EL9 and EL10 RPMs.

3.2 Enable the required repositories

On AlmaLinux or Rocky Linux:

```
sudo dnf install -y epel-release
sudo dnf config-manager --set-enabled crb
```

On RHEL, enable the repositories required by your organization before installing the packages. Repository names and subscription commands can differ from AlmaLinux and Rocky Linux.

```
RHEL_MAJOR="$(rpm -E %rhel)"
ARCH="$(uname -m)"
sudo subscription-manager repos \
  --enable "codeready-builder-for-rhel-${RHEL_MAJOR}-${ARCH}-rpms"
sudo dnf install -y \
  "https://dl.fedoraproject.org/pub/epel/epel-release-latest-${RHEL_MAJOR}.noarch.rpm"
```

3.3 Install MDSPACE on Enterprise Linux 9

Install the EL9 external dependency RPMs directly from the continuous GitHub release, then install the matching local MDSPACE RPM. dnf will also resolve dependencies available from the enabled EL9 repositories.

```
ARCH="$(uname -m)"
case "$ARCH" in
  x86_64|aarch64) ;;
  *) echo "Unsupported architecture: $ARCH" >&2; exit 1 ;;
esac
BASE="https://github.com/MDSPACE-toolkit/mdspace-external-rpms/releases/download/continuous"
sudo dnf install -y \
  "$BASE/elnemo-1.0.0-1.el9.${ARCH}.rpm" \
  "$BASE/genesis-2.1.6.2-0.el9.${ARCH}.rpm" \
  "$BASE/smog2-2.5-5.el9.${ARCH}.rpm" \
  "$BASE/vtk-qt6-lean-9.6.1-1.el9.${ARCH}.rpm" \
  "$BASE/xmipp-3.25.06.0-4.el9.${ARCH}.rpm" \
  ./mdspace-<version>-<release>.el9.${ARCH}.rpm
```

Replace the MDSPACE placeholder filename with the name of the supplied RPM. Install the architecture-specific binary RPMs; the .src.rpm files are source packages and are not required for a normal installation.

Release assets: <https://github.com/MDSPACE-toolkit/mdspace-external-rpms/releases/tag/continuous>. The release also provides SHA256SUMS files for integrity verification.

3.4 Install MDSPACE on Enterprise Linux 10

Install the EL10 external dependency RPMs directly from the same continuous GitHub release, then install the matching EL10 MDSPACE RPM:

```
ARCH="$(uname -m)"
case "$ARCH" in
  x86_64|aarch64) ;;
  *) echo "Unsupported architecture: $ARCH" >&2; exit 1 ;;
esac
BASE="https://github.com/MDSPACE-toolkit/mdspace-external-rpms/releases/download/continuous"
sudo dnf install -y \
  "$BASE/elnero-1.0.0-1.el10.${ARCH}.rpm" \
  "$BASE/genesis-2.1.6.2-0.el10.${ARCH}.rpm" \
  "$BASE/smog2-2.5-5.el10.${ARCH}.rpm" \
  "$BASE/vtk-qt6-lean-9.6.1-1.el10.${ARCH}.rpm" \
  "$BASE/xmipp-3.25.06.0-3.el10.${ARCH}.rpm" \
  ./mdspace-<version>-<release>.el10.${ARCH}.rpm
```

Do not mix EL9 and EL10 RPMs. Use x86_64 assets on x86_64 systems and aarch64 assets on aarch64 systems.

3.5 Launch and verify MDSPACE

```
mdspace
```

3.6 Updating or removing the native installation

To update MDSPACE or its external tools, rerun dnf install with the newer matching RPM assets. To inspect installed MDSPACE-related packages:

```
rpm -qa | grep -Ei 'mdspace|elnero|genesis|smog2|xmipp|vtk-qt6-lean'
```

Remove only packages that you intentionally installed for MDSPACE, for example:

```
sudo dnf remove mdspace
```

4. Portable Linux bundle

The portable bundle is the recommended method for testing MDSPACE. It contains MDSPACE together with the required external command-line programs, does not require system-wide installation, and is intended to work across Linux distributions that provide a sufficiently recent glibc and a graphical desktop environment.

4.1 Check the installed glibc version

```
getconf GNU_LIBC_VERSION
# Alternative:
ldd --version | head -n 1
```

The first command normally prints a value such as glibc 2.34 or glibc 2.39.

Installed glibc	Bundle to use	Compatibility
Older than 2.34	Virtual machine	The portable Linux bundles are not

		supported.
2.34 to 2.38	EL9 bundle	Built on the EL9 glibc 2.34 baseline.
2.39 or newer	EL10 bundle	Built on the EL10 glibc 2.39 baseline.

Why this matters: Linux binaries can require glibc symbols introduced by the system on which they were built. EL9 uses the glibc 2.34 baseline, while EL10 uses glibc 2.39. A bundle cannot run on a system whose glibc is older than the bundle requires.

4.2 System runtime requirements

The portable MDSPACE bundle contains MDSPACE and its external command-line tools, but it does not package the system OpenMPI runtime. OpenMPI and the GCC OpenMP runtime must therefore be installed on the host system before launching the bundle.

Ubuntu and Debian-based distributions

```
sudo apt update
sudo apt install -y \
  openmpi-bin \
  libopenmpi3 \
  libgomp1
```

Enterprise Linux-based distributions

This applies to AlmaLinux, Rocky Linux, Red Hat Enterprise Linux, and compatible distributions:

```
sudo dnf install -y \
  openmpi \
  libgomp
```

Arch Linux-based distributions

```
sudo pacman -Syu --needed \
  openmpi \
  gcc-libs
```

OpenMPI provides the MPI runtime required for parallel GENESIS execution, while libgomp provides the GNU OpenMP runtime used by OpenMP-enabled components.

4.3 Download the correct bundle

All bundles are available at <https://gallois.cc/assets/cryonet.html>. Choose both the EL generation and CPU architecture:

System	x86_64 asset	aarch64 asset
glibc 2.34–2.38	MdSpace-bundle-el9-x86_64.run	MdSpace-bundle-el9-aarch64.run
glibc 2.39 or newer	MdSpace-bundle-el10-x86_64.run	MdSpace-bundle-el10-aarch64.run

4.4 Run the bundle

The bundle is a self-extracting executable. It can be launched directly without administrator privileges:

```
chmod +x MdSpace-bundle-el9-x86_64.run
./MdSpace-bundle-el9-x86_64.run
```

Replace the filename with the EL10 or aarch64 variant selected above. Keep the file in a directory where you have write permission because temporary runtime data may be created there or under the system temporary directory.

4.5 Troubleshooting portable bundles

Message or symptom	Action
GLIBC_x.y not found	The selected bundle is newer than the host glibc. Use the EL9 bundle when possible, or use the virtual machine.
Exec format error	The CPU architecture is wrong. Check <code>uname -m</code> and download <code>x86_64</code> or <code>aarch64</code> accordingly.
Permission denied	Run <code>chmod +x</code> on the <code>.run</code> file and ensure the filesystem permits executable files.
No display or Qt platform error	Run from a graphical Linux session. Remote systems need X11 forwarding, VNC, or another working graphical session. Select the appropriate bundle version.

5. Windows and macOS: run the bundle in an AlmaLinux 9 virtual machine

Native MDSPACE packages are not currently distributed for Windows or macOS. The recommended testing setup is to create an AlmaLinux 9 virtual machine, then download and run the EL9 portable bundle inside it. Prebuilt AlmaLinux 9 virtual machines are also supplied and can be imported instead. All assets are available at <https://gallois.cc/assets/cryonet.html>.

CPU compatibility: Use an x86_64 AlmaLinux image on Intel or AMD hosts and an aarch64 image on ARM64 hosts. Intel Macs require x86_64, while Apple Silicon Macs require aarch64. Matching the guest and host architectures provides native virtualization performance.

5.1 Choose the virtualization software

Use virtualization software appropriate for the host operating system. The official AlmaLinux installation guide is available at <https://wiki.almalinux.org/documentation/installation-guide.html>.

Windows

1. Install VirtualBox or another virtualization application suitable for your computer.
2. Download the AlmaLinux 9 installation ISO matching the host CPU architecture.
3. Create a new Linux virtual machine, attach the AlmaLinux 9 ISO, and install AlmaLinux.
VirtualBox documentation: <https://www.virtualbox.org/manual/>.
4. Allocate the recommended CPU, memory, storage, and display resources, and select a graphical AlmaLinux installation.

macOS

5. Install UTM on the Mac. UTM installation documentation: <https://docs.getutm.app/installation/macos/>.
6. Download the AlmaLinux 9 installation ISO matching the Mac CPU architecture: `x86_64` for Intel or `aarch64` for Apple Silicon.
7. Create a new Linux virtual machine in UTM, attach the AlmaLinux 9 ISO, and install AlmaLinux.
8. Review the virtual-machine resources and select a graphical AlmaLinux installation.

5.2 Create a new AlmaLinux 9 virtual machine (recommended)

Create the virtual machine using the host-appropriate software and an AlmaLinux 9 ISO. The following combinations are recommended:

Host platform	CPU architecture	Recommended setup
Windows	x86_64	VirtualBox with an AlmaLinux 9 x86_64 ISO
Windows	aarch64	ARM64-compatible virtualization with an AlmaLinux 9 aarch64 ISO
macOS (Intel)	x86_64	UTM with an AlmaLinux 9 x86_64 ISO
macOS (Apple Silicon)	aarch64	UTM with an AlmaLinux 9 aarch64 ISO

5.3 Alternative: import a prebuilt virtual machine

- Prebuilt VM assets are available at <https://gallois.cc/assets/cryonet.html>.
- On Windows, create a new virtual machine and attach the supplied VMDK as an existing virtual disk.
- On macOS, open the supplied .utm package in UTM.
- Choose the prebuilt VM matching the host CPU architecture, then review its CPU, memory, display, and storage settings.
- Save or import the machine, then start it. The prebuilt VM already contains the required software.

5.4 Recommended virtual-machine resources

Resource	Minimum	Recommended
CPU	2 virtual CPUs	4 or more virtual CPUs
Memory	16 GB	32 GB or more
Free disk space	30 GB	50 GB or more
Display	128 MB video memory	Maximum supported video memory

Do not allocate all host CPU cores or all host memory to the VM. Leave enough resources for Windows or macOS to remain responsive.

5.5 Install and run the bundle in the virtual machine

- Start the AlmaLinux 9 virtual machine and complete the graphical installation when using a newly created VM.
- Open a terminal in AlmaLinux 9.
- Install the runtime requirements with: `sudo dnf install -y openmpi libgomp`
- Download the EL9 bundle matching the guest CPU architecture from <https://gallois.cc/assets/cryonet.html>.
- Make the bundle executable and run it as described in Section 4.4. For a prebuilt VM, use the supplied login credentials and launch MDSPACE from the application menu or desktop shortcut.

5.6 Sharing files with the virtual machine

The most portable methods are cloud storage, or a network transfer. Shared folders may also be configured when supported by the selected virtualization application:

- Power off the virtual machine.
- Open the virtual-machine settings and configure a shared host directory.
- Enable the relevant guest integration or shared-folder option when required.
- Start the virtual machine and open the shared directory from AlmaLinux.

Guest integration: Shared folders, dynamic display resizing, clipboard integration, and improved pointer handling may require the guest tools supplied by the selected virtualization application. The distributed virtual machines should include the appropriate integration packages when available.

5.7 Shut down correctly

Use the AlmaLinux desktop shutdown command before closing the virtualization application. Avoid forcing the virtual machine off repeatedly, because this is equivalent to disconnecting power from a physical computer and can corrupt the virtual disk.